Lab Note: Week 2 Friday, September, 5, 2025

Creating a Git Repository and Uploading a File

Learning Objectives

- Understand how to initialize a Git repository locally.
- Learn to track files using git add and commit changes.
- Connect to a remote repository (e.g., GitHub) and push code.
- Understand basic Git workflow from initialization to remote push.

Step-by-Step Instructions

1. Before You Begin

- Download and install Git if not installed.
- Create a GitHub account (if you don't have one)

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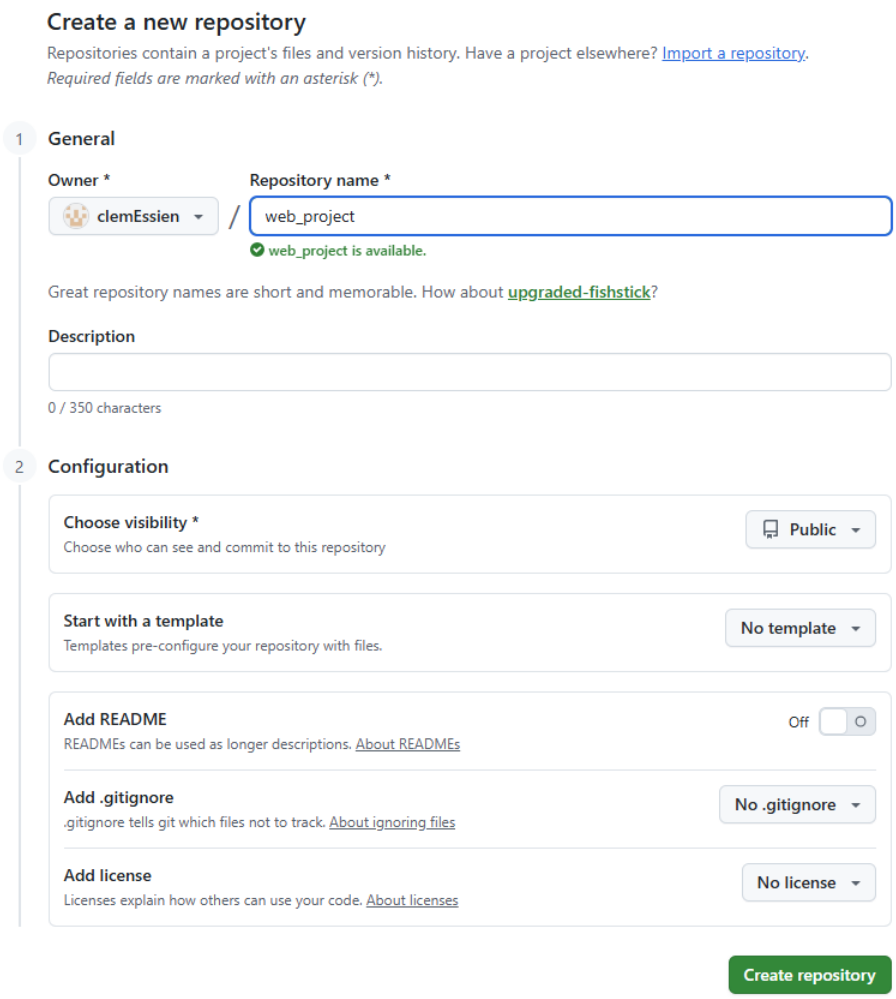
2. Create a new repository

- Login to your GitHub account
- Click on New to create a repository



- Give the project a name of your choice
- Ensure that visibility is set to public
- Click on create repository

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2. Initializing a Local Repository

- Navigate to your project folder - **web_app** (or create one if it doesn't exist)
  - mkdir **web_app**
  - cd **web_app**
- Initialize Git in this folder by running the command below
  - **git init**

This sets up a hidden .git folder, marking your project as a Git repository

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3. Create and Stage a File

- Create a file called readme.md in app folder i.e. **touch** README.md
  (Make sure the file extension is .md)
- Check the repository status by running the command below:
  - **git status**
- Stage the new file for commit by running
  - **git add** README.md

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4. Commit Your Changes

- Commit changes to the repository
  - git commit -m "Adding README.md"
- create a new branch
  - git branch -M main
- Add remote repository to your local repository
  - git remote add origin https://github.com/**YOUR_USERNAM**E/web_app.git
- push your main branch to the remote repository
  - git push -u origin main
    when prompted for a password, use your GitHub Personal Access Token in Step 5.

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5. GitHub Authentication (Important)

Note: GitHub no longer supports passwords for Git operations.
You must use a **Personal Access Token (PAT)** instead.

Step A: Create a Personal Access Token (One-Time Setup)

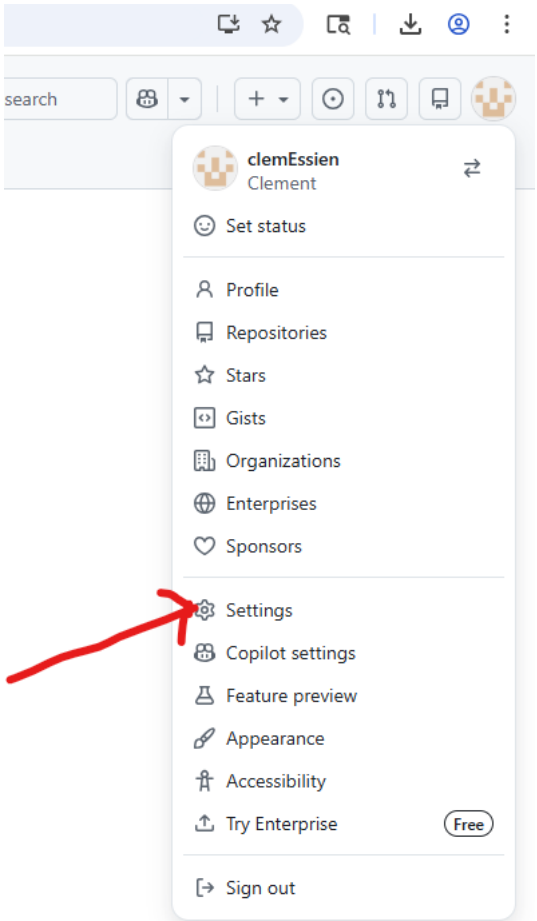
1. Log in to **GitHub**
2. Click your profile picture (top-right) → **Settings**
3. Click **Developer settings**
4. Click **Personal access tokens** → **Tokens (classic)**
5. Click **Generate new token (classic)**

Fill in:

- **Note:** CSIS Git Access
- **Expiration:** 90 days (recommended)
- **Scopes:** ☑ repo

Click **Generate token**

⚠ **IMPORTANT:**
Copy the token immediately. You will NOT be able to see it again.


search
clemEssien
Clement
Set status
Profile
Repositories
Stars
Gists
Organizations
Enterprises
Sponsors
Settings
Copilot settings
Feature preview
Appearance
Accessibility
Try Enterprise
Free
Sign out

Clement (clemEssien)
Your personal account

Go to your personal profile

- Public profile
- Account
- Appearance
- Accessibility
- Notifications

Access

- Billing and licensing
- Emails
- Password and authentication
- Sessions
- SSH and GPG keys
- Organizations
- Enterprises
- Teams
- Moderation

Code, planning, and automation

- Repositories
- Codespaces
- Models (Preview)
- Packages
- Copilot
- Pages
- Saved replies

Security

- Code security

Integrations

- Applications
- Scheduled reminders

Archives

- Security log
- Sponsorship log

- Developer settings

Public profile

Name

Clement

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your email settings

Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Pronouns

Don't specify

URL

Social accounts

Link to social profile 1

Link to social profile 2

Link to social profile 3

Link to social profile 4

Company

You can @mention your company's GitHub organization to link it.

Location

Display current local time

Other users will see the time difference from their local time.

ORCID iD

ORCID provides a persistent identifier - an ORCID iD - that distinguishes you from other researchers. Learn more at ORCID.org.

Connect your ORCID iD

All of the fields on this page are optional and can be deleted at any time, and by filling them out, you're giving us consent to share this data wherever your user profile appears. Please see our privacy statement to learn more about how we use this information.

Update profile

Profile picture

Edit

Contributions & activity

github.com/settings/tokens

Settings / Developer Settings

- GitHub Apps
- OAuth Apps
- Personal access tokens
  - Fine-grained tokens
  - Tokens (classic)

Personal access tokens (classic)

No personal access token created

Need an API token for scripts or testing? Generate a personal access token for quick access to the GitHub API.

Generate new token

GitHub API

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to authenticate to the API over Basic Authentication.

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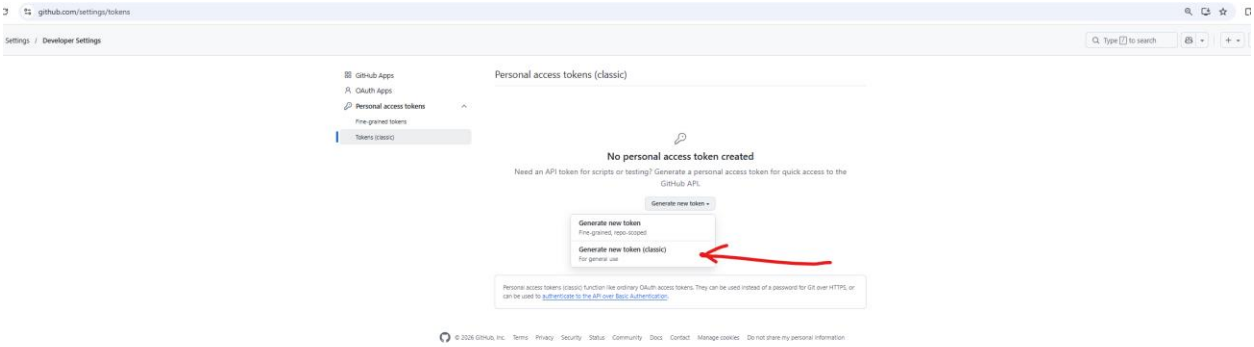

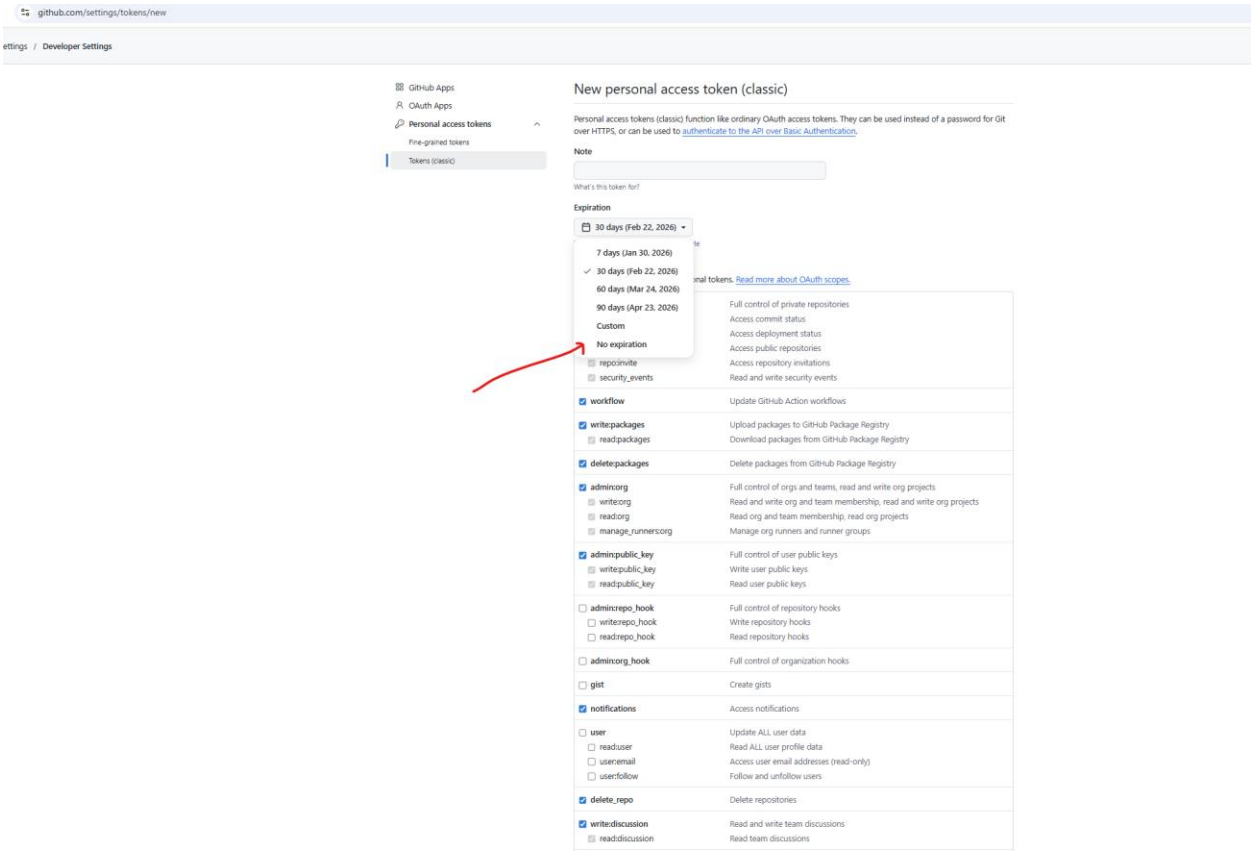

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6. Subsequently as you work on your project

The only commands you will need to run each time will be:

- **git status** – To view list of files in your project that have been modified
- **git add .** – to all new / modified files
- or **git add specific_file_name** - to add a specific file name
- **git commit -m** "meaning message on what you worked on or what changes you made"
- **git push – to push code changes to the remote branch**

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7. Verify on GitHub

Congratulations—you've successfully created your repository.